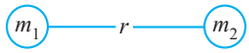
8 Motion of System of Particles & C.O.M

Location of C.O.M

1. Find Location of Centre of Mass from mass 1 and 2



2. The centre of mass of a system of particles does not depend on

(*a*) Masses of the particles
(*b*) Internal forces of the particles
(*c*) Position of the particles
(*d*) Relative distance between the particles

3. Position of centre of mass of a body:

(*a*) Depends on the choice of co-ordinate system
(*b*) Is independent of the choice of co-ordinate system
(*c*) May or may not depend on the choice of co-ordinate system
(*d*) None of the above

4. A system of consists of mass M and m ($<<M$). The centre of mass of the system is

(*a*) At the middle (*b*) Nearer to M
(*c*) Nearer to m (*d*) the position of larger mass

5. The centre of mass of a system of two particles divides the distance between them

(*a*) In inverse ratio of square of masses of particles
(*b*) In direct ratio of square of masses of particles
(*c*) In inverse ratio of masses of particles
(*d*) In direct ratio of masses of particles

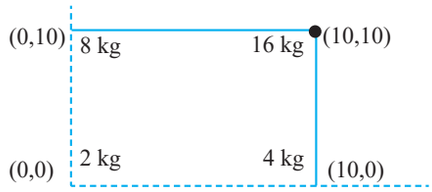
6. The centre of mass of a body

(*a*) Lies always outside the body
(*b*) May lie within, outside the surface of the body
(*c*) Lies always inside the body
(*d*) Lies always on the surface of the body

7. A point object of mass m is kept at (a, 0) along x-axis. What mass should be kept at (-3 a, 0), so that centre of mass lies at origin.

(*a*) m (*b*) 2 m
(*c*) $m/3$ (*d*) 3 m

8. Find position of C.O.M.?



(*a*) $\frac{20}{3}, 8$ (*b*) $8, \frac{20}{3}$ (*c*) $\frac{20}{3}, \frac{20}{3}$ (*d*) 8,4

9. n particles of masses m,2 m, 3 m... 4 m grams are placed on the same line at distances l, $2l$, $3l$, ... $4l$ c.m. from a fixed point. The distance of centre of mass of the particles from the fixed point in centimeter is

(*a*) $\frac{(2n+1)l}{3}$ (*b*) $\frac{l}{n}+l$

(*c*) $\frac{n(n^2+1)l}{2}$ (*d*) $\frac{l}{n}-l$

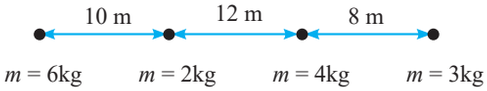
10. The position vector of three particle of mass m_1=3 kg, m_2=2 kg, m_3 = 5 kg are $(\hat{i},\hat{j},\hat{k})$, $(2\hat{i}+\hat{j}+\hat{k})$ and $(\hat{i}-\hat{j}+\hat{k})$

(*a*) $1.2\hat{i}+\hat{k}$ (*b*) $\hat{i}+1.2\hat{j}+3\hat{k}$

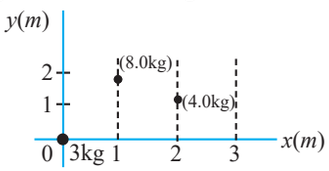
(*c*) $2\hat{i}+3\hat{k}+\hat{j}$ (*d*) $4\hat{j}+1.2\hat{k}$

11. Two bodies of masses 1 kg and 2kg are lying in xy plane at (-1, 2) and (2, 4) respectively. What are the coordinates of the centre of mass?

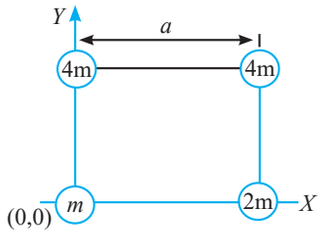
12. What is the distance of C.O.M from 4 kg mass in the following arrangement



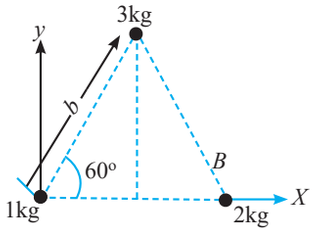
13. What are the coordinates of the centre of mass of the three particles system shown in figure?



14. Four particles of masses *m*, 2m, 3m, 4m are placed at the corners of a square of side '*a*' as shown in fig. Find out the co-ordinates of centre of mass.



15. Locate the centre of mass of a system of particles of masses 1kg, 2 kg and 3 kg situated at the corners of an equilateral triangle of side b as shown in figure.

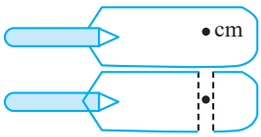


16. The coordinates of a triangle *ABC* are *A*(1, 2), *B*(4, 6) and C(–3, –2). Three particles of masses 1kg, 2 kg and m kg are placed the vertices of the triangle. If the coordinates of the centre of mass are $\left(\frac{3}{5},2\right)$, calculate the mass *m*.

17. Three identical spheres each of mass *M* are placed at corners of right triangle with mutually perpendicular sides equal to 3m each. Taking point of intersection of mutually perpendicular sides as origin, the magnitude of position vector of centre of mass of the system will be $\sqrt{x}\ m$. The value of *x* is

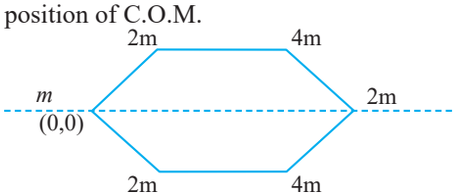
18. Find the position vector of centre of mass of a system of three particles of masses 1kg, 2kg and 3kg located at position vectors $\vec{r_1}=(4\hat{i}+2\hat{j}-3\hat{k})m, \vec{r_2}=(\hat{i}-4\hat{j}+2\hat{k})m$ and $\vec{r_3}=\left(2\hat{i}-2\hat{j}+\hat{k}\right)$ respectively.

19. A cricket bat is cut at the location of its centre of mass as shown. Then



(*a*) The two pieces will have the same mass
(*b*) The bottom piece will have large mass
(*c*) The handle piece will have larger mass
(*d*) Mass of handle piece is double the mass of bottom piece

20. Find position of C.O.M.



21. Two bodies of mass 1 kg and 3 kg have position vectors $\hat{i}+2\hat{j}+\hat{k}$ and $-3\hat{i}-2\hat{j}+\hat{k}$ respectively. The magnitude of position vector of centre of mass of this system will be similar to the magnitude of vector:

(*a*) $\hat{i}-2\hat{j}+\hat{k}$ (*b*) $-3\hat{i}-2\hat{j}+\hat{k}$
(*c*) $-2\hat{i}+2\hat{k}$ (*d*) $-2\hat{i}-\hat{j}+\hat{k}$

C.O.M of Continuous Mass System

22. A rod of length L placed along *x*-axis having linear mass density is proportional to *x* then distance of COM from one end.

(*a*) L/2 (*b*) 2L/3
(*c*) L/4 (*d*) 3L/2

23. Mass is non-uniformly distributed on the circumference of a ring of radius *r* and centre at origin. Let a be the distance of the centre of mass of the ring form origin. Then

(*a*) $r = a$ (*b*) $0 \le a \le r$
(*c*) $a < r$ (*d*) $a > r$

24. The center of mass of a solid hemisphere of radius 8 cm is *x* cm form the centre of the flat surface. The value of *x* is

25. If linear density of a rod of length 3m varies as $\lambda = 2 + x$, then the position of the center of gravity of the rod is

(*a*) $\frac{7}{3}m$ (*b*) $\frac{12}{7}m$
(*c*) $\frac{10}{7}m$ (*d*) $\frac{9}{7}m$

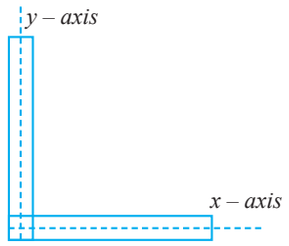
26. Length of a rod is 3m and its mass acting per unit length is directly proportional to distance *x* from one of its end then its centre of gravity from that end will be at :-

(*a*) 1.5*m* (*b*) 2 m
(*c*) 2.5m (*d*) 3m

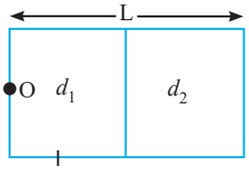
27. The linear mass density (λ) of a rod of length *L* kept along *x*-axis varies as $\lambda = \alpha + \beta x$; where α and β are positive constants. The centre of mass of the rod is at

(*a*) $\frac{(2\beta+3\alpha L)L}{2(2\beta+\alpha L)}$ (*b*) $\frac{(3\alpha+2\beta L)L}{3(2\alpha+\beta L)}$
(*c*) $\frac{(3\beta+2\alpha L)L}{3(2\beta+\alpha L)}$ (*d*) $\frac{(3\beta+2\alpha L)L}{3\beta+2\alpha)}$

28. Two Identical Rod (M,L) placed along *x* and *y*-axis as shown in fig. Then find Location of C.O.M.



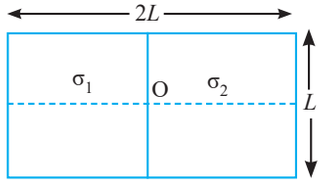
29. Shows a rectangular plane of length L, the half of which is made of material of density d_1 and another half of density d_2. The distance of centre of mass from the origin O is



(a) $\frac{d_1+2d_2}{2(d_1+d_2)}L$ (b) $\frac{(d_1+3d_2)L}{4(d_1+d_2)}$

(c) $\frac{(d_1+3d_2)L}{2(d_1+d_2)}$ (d) $\frac{(3d_1+d_2)L}{4(d_1+d_2)}$

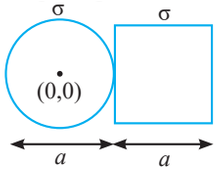
30. Find location of centre of mass of system



(a) $\left(\frac{\sigma_1+3\sigma_2}{\sigma_1+\sigma_2}\right)\frac{L}{2}$ (b) $\left(\frac{3\sigma_1+\sigma_2}{\sigma_1+\sigma_2}\right)\frac{L}{2}$

(c) $\left(\frac{\sigma_1-3\sigma_2}{\sigma_1+\sigma_2}\right)L$ (d) $\left(\frac{\sigma_1+\sigma_2}{\sigma_1+3\sigma_2}\right)\frac{L}{2}$

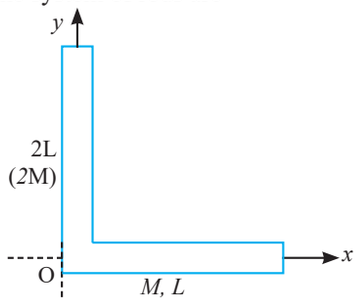
31. Find centre of mass from origin for a given combination of circular plate and square plate of same density (σ).



(a) $\frac{4a}{\pi+4}$ (b) $\frac{4a}{\pi}$

(c) $\frac{a}{\pi+4}$ (d) $\frac{a}{4}$

32. Figure shows a composite system of two uniform rods of lengths as indicated. Then the coordinates of the centre of mass of the system of rods are



(a) $\left(\frac{L}{2},\frac{2L}{3}\right)$ (b) $\left(\frac{L}{4},\frac{2L}{3}\right)$

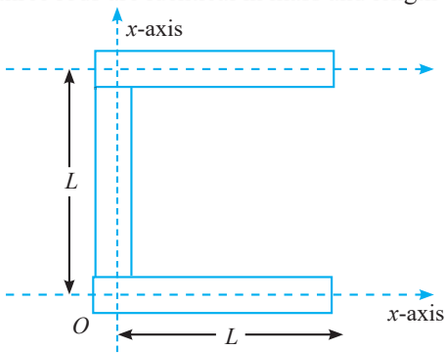
(c) $\left(\frac{L}{6},\frac{2L}{3}\right)$ (d) $\left(\frac{L}{6},\frac{L}{3}\right)$

33. A uniform metre rod is bent into L shape with the bent arms at 90° to each other. The distance of the center of mass from the bent point is

(a) $\frac{L}{4\sqrt{2}}m$ (b) $\frac{L}{2\sqrt{2}}m$

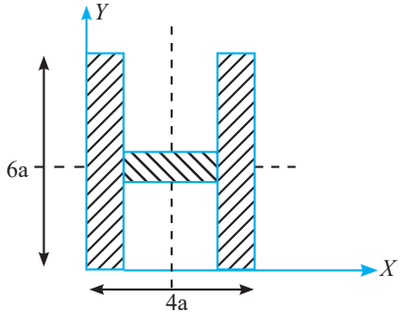
(c) $\frac{L}{\sqrt{2}}m$ (d) $\frac{L}{8\sqrt{2}}m$

34. Locate the centre of mass of arrangement shown in figure. The three rods are identical in mass and length



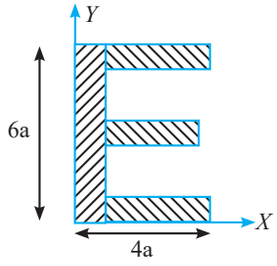
(a) $\left(\frac{L}{2},\frac{L}{2}\right)$ (b) $\left(\frac{L}{3},0\right)$ (c) $\left(\frac{L}{3},\frac{L}{2}\right)$ (d) $\left(0,\frac{L}{3}\right)$

35. Co-ordinate of COM for given object will be-



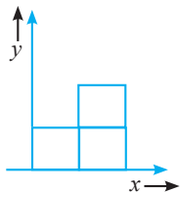
(a) (2a, a) (b) (2a, 3a) (c) (a, 3a) (d) (3a, 3a)

36. Co-ordinate of COM for given object will be-

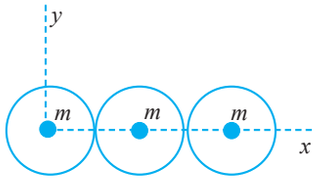


(a) (2a, 3a) (b) (a, 3a) (c) (a, 2a) (d) (a, 4a)

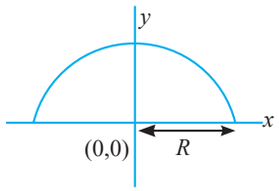
37. Find position of COM for given combination (each plate is square of side a and mass m).



38. Three identical spheres, each of mass M and radius R, are placed in a row as shown in the figure. Find the location of the center of mass (C.O.M.) of the system..



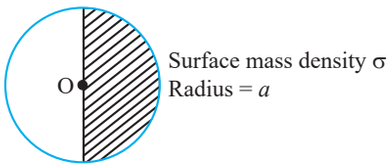
39. Find the position of the center of mass of a half ring, given its radius R. The center of the ring is located at the origin (0,0)



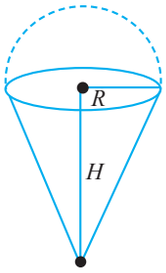
(*a*) $\left(\frac{2\pi}{R}\right)$ (*b*) $\left(\frac{2R}{\pi}\right)$

(*c*) $\left(\frac{R}{2}\right)$ (*d*) $\left(\frac{R}{2\pi}\right)$

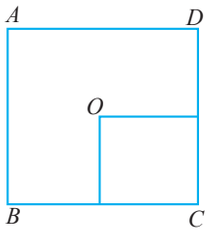
40. The disc of mass M with uniform surface mass density σ is shown in the figure. The centre of mass of shaded area is at the position $\left(\frac{xa}{6\pi},0\right)$ where x is_____



41. Centre of mass of solid cone and solid hemisphere (same density) is at center then find relation between R and H.



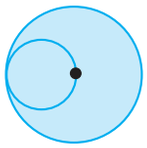
42. From a uniform square plate, one-fourth part is removed as shown. The centre of mass of remaining part will lie on



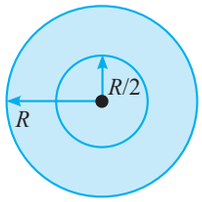
(*a*) OC (*b*) OA

(*c*) OB (*d*) OD

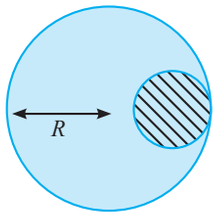
43. From a uniform circular disc of mass M and radius R, a small circular disc of radius $R/2$ is removed in such a way that both have a common tangent. Find the distance of centre of mass of remaining part from the centre of original disc.



44. A disc of mass M and radius R given, a small disc of $R/2$ is removed as shown in figure then find the C.O.M of remaining part.

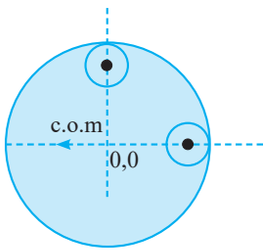


45. A uniform disc of padius ($R/4$) is removed as shown in figure then c.o.m of remaining part.



46. Find the centre of mass of a uniform disc of radius 'a' from which a circular section of radius 'b' has been removed. The centre of the hole is at a distance c from the centre of the disc.

47. A disc of mass (M,R), Two disc of radius $R/4$ is cut and removed as shown in figure then C.O.M of remaining part



48. Blocks A and B kept on a smooth horizontal surface are given equal speeds of 2 m/s in opposite sense as shown in the figure.

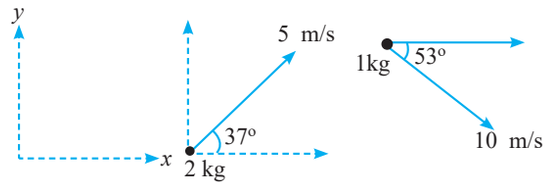


At $t = 0$ the position of blocks are shown, then the coordinates of centre of mass of system of two blocks, at t = 3s will be:

Velocity of Center of Mass

49. Two particles of masses 1 kg and 0.5 kg are moving in the same direction with speeds of 2 m/s and 6 m/s, respectively, on a smooth horizontal surface. Find the speed of the centre of mass of the system

50. Find velocity of COM.



51. Two blocks of masses 5kg and 2kg are connected by a spring of negligible mass and placed on a frictionless horizontal surface. An impulse provides a velocity of 7 m/s to the heavier block in the direction of the lighter block. The velocity of the centre of mass is/

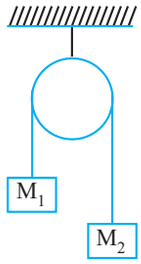
(*a*) 4 m/s (*b*) 5 m/s (*c*) 2 m/s (*d*) 3 m/s

Acceleration of Center of Mass

52. Three particles of masses 1kg, 2 kg and 3kg are subjected to forces $(3\hat{i}-2\hat{j}+2\hat{k})$N,$(-\hat{i}+2\hat{j}-\hat{k})$N and $(\hat{i}+\hat{j}+\hat{k})$N respectively. Find the magnitude of the acceleration of the CM of the system

53. Two particles of mass 1kg and 3kg are having acceleration $(\hat{i}-2\hat{j}+5\hat{k})m/s^2$ and $(-2\hat{i}+\frac{4}{3}\hat{j}-\hat{k})m/s^2$. Find out $\underset{\sim}{a}_{cm}$

54. Find acceleration of centre of mass.



55. A ball of mass m is thrown upward and another ball of same mass is thrown downward so as to move freely under gravity. The acceleration of centre of mass is

(*a*) g (*b*) 9g/3 (*c*) 2g/3 (*d*) g/2

For System of Object

56. Position of C.O.M will only change when external force will act

(*a*) True (*b*) False

57. Momentum of system (C.O.M) will only change when external force will act

(*a*) True (*b*) False

58. K.E. of system will only change when external force will act

(*a*) True (*b*) False

59. Internal force can change K-E of System→ Yes/No

60. Internal force can change Momentum of system→ Yes/No

61. Internal force Can Change Moment of object → Yes/No

62. Internal forces can change

(*a*) The linear momentum but not the kinetic energy

(*b*) The kinetic energy but not the linear momentum

(*c*) Linear momentum as well as kinetic energy

(*d*) Neither the linear momentum nor the kinetic energy

63. Which of the following statement is correct:-

(*a*) Kinetic energy of a system can be changed without changing its momentum.

(*b*) Kinetic energy of a system cannot be changed without changing its momentum.

(*c*) Momentum of a system cannot be changed without changing its kinetic energy.

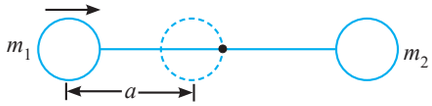
(*d*) Body cannot have energy without having momentum.

64. A child is standing at one end of a long trolley moving with a speed v on a smooth horizontal track. If the child starts running towards the other end of the trolley with a speed u, the centre of mass of the system (trolley + child) will move with a speed

(*a*) Zero (*b*) $(v+u)$ (*c*) v (*d*) $(v-u)$

Shift in Center of Mass

65. Consider a system having two masses m_1 and m_2 in which first mass is pushed towards the centre of mass by a distance a. The distance by which the second mass should be moved to keep the centre of mass at same position



(*a*) $\frac{m_1}{m_2}a$ (*b*) $\frac{m_1}{(m_1+m_2)}a$

(*c*) $\frac{m_2}{m_1}a$ (*d*) $\left(\frac{m_2}{(m_1+m_2)}\right)a$

66. Two blocks of masses 10 kg and 30 kg are placed on the same straight line with coordinates [0, 0] cm and [x, 0] cm respectively. The block of 10 kg is moved on same line through a distance of 6 cm towards the other block. The distance through which block of 30 kg must be moved to keep the position of C.O.M. of system unchanged is:

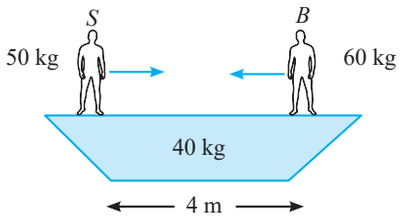
(*a*) 4 cm towards the 10 kg block

(*b*) 2 cm away from the 10 kg block

(*c*) 2 cm towards the 10kg block

(*d*) 4 cm away from the 10 kg block

67. Santi(50kg)andBanti(60kg)aresittingatthetwoextremesofa 4 *m* long boat (40 kg) standing still in water, To discuss a mechanics problem, they come to the middle of the boat. Neglecting friction with water, how far does the boat move on the water during the process?



Conservation of Momentum for System of Particles

68. A man of mass *m* is suspended in air by holding the rope of a balloon of mass *M*. As the man climbs up the rope, the balloon

(*a*) Moves upward (*b*) Moves downward

(*c*) Remains stationary (*d*) Can not Say

69. Two particles *A* and *B*, initially at rest moves towards each other under the mutual force of attraction. At the instant when speed of *A* is *v* and speed of *B* is 2v, the speed of centre of mass of the system is:

(*a*) 3v (*b*) *v* (*c*) 1.5v (*d*) Zero

70. A man (mass = 50 kg) and his son (mass = 20 kg) are standing on a frictionless surface facing each other. The man pushes his son so that he starts moving at speed of 0.7 ms^{-1} with respect to man. The speed of man with respect to surface is :

(*a*) 0.25 ms^{-1} (*b*) 0.20 ms^{-1} (*c*) 0.47 ms^{-1} (*d*) 0.14 ms^{-1}

71. A man of half the weight of trolley car is running on the road and suddenly jumps into a stationary trolley car. Then, the whole system of trolley car along with man weighing 180 kg starts moving with velocity 2 m/s. The velocity of the running man was m/s, when he jumps into the car

72. A man of 50 kg mass is standing in a gravity free space at a height of 10 m above the floor. He throws a stone of 0.5 kg mass downwards with a speed 2 m/s. When the stone reaches the floor, the distance of the man above the floor will be

(*a*) 9.9 m (*b*) 10.1m (*c*) 10 m (*d*) 20 m

73. Two spherical bodies of mass *M* and 5*M* and radii *R* and 2*R* respectively are released in free space with initial separation between their centres equal to 12*R*. If they attract each other due to gravitational force only, then the distance covered by the smaller body just before collision is

(*a*) 1.5 R (*b*) 2.5 R (*c*) 4.5 R (*d*) 7.5 R

74. A ball strikes the floor and after collision rebounds back. In this state

(*a*) Momentum of the ball is conserved

(*b*) Mechanical energy of the ball is conserved

(*c*) Momentum of ball-earth system is conserved

(*d*) The kinetic energy of ball-earth system is conserved.

75. A mass of 16 kg moving with a speed of 5 m/s collides with another stationary mass of 4 kg. As a result of the collision, the two masses stick together. The kinetic energy of the composite mass will be: -

(*a*) 200 J (*b*) 160 J (*c*) 320 J (*d*) 128 J

76. A bullet of mass *m* is fired from a gun of mass *M*. The recoiling gun compresses a spring of force constant *k* by a distance *d*. Then the velocity of the bullet is :-

(*a*) $kd\sqrt{M/m}$ (*b*) $\frac{d}{M}\sqrt{km}$

(*c*) $\left(\frac{d}{m}\right)\sqrt{kM}$ (*d*) $\frac{kM}{m}\sqrt{d}$

77. An artillery piece of mass m_1 fires a shell of mass m_2 horizontally after the firing, the ratio of kinetic energy of the artillery and that of shell is:

(*a*) $\frac{m_1}{m_1+m_2}$ (*b*) $\frac{m_2}{m_1}$ (*c*) $\frac{m_1}{m_2}$ (*d*) $\frac{m_2}{m_1+m_2}$

Explosion

78. A shell following a parabolic path explodes somewhere in its flight due to some internal forces. The centre of mass of fragments will move in

(*a*) Vertical direction (*b*) Any direction

(*c*) Horizontal direction (*d*) Same parabolic path

79. A body at rest explodes into two pieces of unequal mass. The part will move

(*a*) Arbitrarily

(*b*) In the same direction

(*c*) In opposite direction with equal speeds

(*d*) In opposite direction with unequal speeds.

80. A ball of mass *M* moving with velocity *v*, suddenly break into two part of rati 2 : 3. If smaller mass comes to at rest then velocity of other mass.

81. A body falling vertically downwards under gravity breaks in two parts of unequal masses. The centre of mass of the two parts together

(*a*) Shifts horizontally towards heavier piece

(*b*) Shifts horizontally towards light pieces

(*c*) Doesn't shift horizontally

(*d*) Shifts horizontally if initial speed is zero

82. A stationary particle explode into two part in mass m_1 & m_2 then Ratio of their K.E.??

83. A body of mass 3m at rest explodes into three-identical pieces. Two of the pieces move with a speed *v* each in mutually perpendicular directions. The total kinetic energy released is: -

84. A bomb initially at rest explodes by itself into three equal mass fragments. The velocities of two fragments are $(2\hat{i}+3\hat{j})$ m/s and $(\hat{i}-2\hat{j})$ m/s. The velocity of the third fragment is

85. A bomb of mass 12 kg explodes into two pieces of 4 kg and 8 kg. The velocity of 4 kg piece is 24 m/s. The kinetic energy of 8 kg piece is :-

(*a*) 192 J (*b*) 786 J (*c*) 576 J (*d*) 687 J

86. A bomb of mass m = 1 kg thrown vertically upwards with a speed u = 100 m/s explodes into two parts after t = 5*s*. *A* fragment of mass m_1 = 400g moves downwards with a speed v_1 = 25 m/s then speed v_2 and direction of another mass m_2 will be :-

(*a*) 40 m/s downwards

(*b*) 40 m/s upwards

(*c*) 60 m/s upwards

(*d*) 100 m/s upwards

87. A 1kg stationary bomb is exploded in three parts having mass ratio 4 :4:7. Parts having same mass move in perpendicular directions with velocity 70 m/s, then the velocity of bigger part will be :-

(*a*) $40\sqrt{2}$m/s (*b*) $40/\sqrt{2}$m/s

(*c*) $20\sqrt{2}$m/s (*d*) $20/\sqrt{2}$m/s

88. A cannon ball is fired with a velocity 200 m/s at an angle of 60° with the horizontal. At the highest point of its flight, it explodes into 3 equal fragments, one going vertically upwards with a velocity 100 m/s, the second one falling vertically downwards with a velocity 100 m/s. The third fragment will be moving with a velocity;

(*a*) 100 m/s in the horizontal direction

(*b*) 300 m/s in the horizontal direction

(*c*) 300 m/s in a direction making an angle of 60° with the horizontal

(*d*) 200 m/s in a direction making an angle of 60° with the horizontal

89. A projectile is projected at a speed u at an angle θ with horizontal. At the highest point projectile splits into two fragments of mass ratio 1 : 2. The smaller fragment coming to rest. Then consider following statements:-

(A) In a given time both parts will cover equal vertical displacement.

(B) COM of two fragments will follow the same path as the object would have followed if not exploded.

(C) Linear momentum is conserved in vertical direction only.

Which of the following is correct?

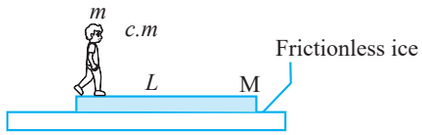
(*a*) A, B, C (*b*) A, B (*c*) B, C (*d*) A, C

90. A shell of mass 200 gm is ejected from a gun of mass 4 kg by an explosion that generates 1.05 kJ of energy. The initial velocity of the shell is

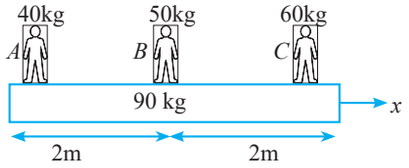
(*a*) 40 ms^{-1} (*b*) 120 ms^{-1} (*c*) 102 ms^{-1} (*d*) 80 ms^{-1}

Man-Plank Problems

91. A man of mass m stand at the left end of a uniform plank of length L and mass M, which lies on a frictionless surface of ice. If the man walks to the other end of the plank, then by what distance does the plank slide on the ice?



92. Three men, *A, B*, and *C*, each with masses of 40 kg, 50 kg, and 60 kg respectively, are standing on a plank of mass 90 kg, placed on a smooth horizontal surface. If *A* and *C* exchange their positions, what will be the shift in the position of man *B*.



Collision

93. Internal force can change velocity of centre of mass

(*a*) True (*b*) False

94. If external force is zero then location of COM will not change

(*a*) True (*b*) False

95. Internal force can change momentum of COM

(*a*) True (*b*) False

96. Internal force can charge K.E. of the COM

(*a*) True (*b*) False

97. Centre of mass is a point who divide system in two equal mass

(*a*) True (*b*) False

98. In every collision momentum is conserved all the time.

(*a*) True (*b*) False

99. In elastic collision K.E. is conserved all the time.

(*a*) True (*b*) False

100. In elastic collision of two object of same mass, speed will interchange

(*a*) True (*b*) False

101. In ball-earth collision, momentum of ball remains conserved

(*a*) True (*b*) False

102. A batsman hits back a ball of mass 0.4 kg straight in the direction of the bowler without changing its initial kinetic energy of 45 joule. The impulse imparted to the ball is –Ns

103. A ball of mass m moving with velocity v collides head-on with the second ball of mass m at rest. If the coefficient of restitution is e and velocity of first ball after collision is v_1 and velocity of second ball after collision is v_2 then

(a) $v_1 = \frac{(1-e)u}{2}, v_2 = \frac{(1+e)u}{2}$

(b) $v_1 = \frac{(1+e)u}{2}, v_2 = \frac{(1-e)u}{2}$

(c) $v_1 = \frac{u}{2}, v_2 = -\frac{u}{2}$

(d) $v_1 = (1+e)\,u,\ v_2 = (1-e)u$

104. Two particles of masses m_1 and m_2 move with initial velocities u_1 and u_2. On collision, one of the particles get excited to higher level, after absorbing energy e. If final velocities of particles be v_1 and v_2 then we must have

(a) $\frac{1}{2}m_1^2u_1^2 + \frac{1}{2}m_2^2u_2^2 + \varepsilon = \frac{1}{2}m_1^2v_1^2 + \frac{1}{2}m_2^2v_2^2$

(b) $m_1^2u_1 + m_2^2u_2 - \varepsilon = m_1^2v_1 + m_2^2v_2$

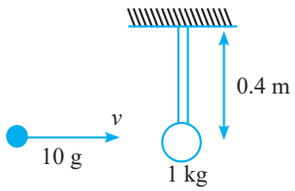
(c) $\frac{1}{2}m_1u_1^2 + \frac{1}{2}m_2u_2^2 = \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2 - \varepsilon$

(d) $\frac{1}{2}m_1u_1^2 + \frac{1}{2}m_2u_2^2 - \varepsilon = \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2$

105. A ball of mass 4 kg moving on a smooth horizontal surface makes an elastic collision with another ball of mass m at rest in the line of motion of first ball. If after collision first ball moves in the same direction with one fourth of its velocity collision, then mass of second ball is

(a) 4 kg (b) 4.4 kg (c) 2.4 kg (d) 2 kg

106. A bullet of 10g, moving with velocity v, collides head-on with the stationary bob attached to massless rod and recoils with velocity 100 m/s. The length of the rod is 0.4 m and mass of the bob is 1 kg. The minimum value of v = ... m/s so that pendulum describes a circle



107. A heavy vehicle moving with velocity 15 m/s strikes an object of very small mass at rest head on elasticity. Velocity of object after collision is

(a) 15 m/s (b) 25 m/s (c) 20 m/s (d) 30 m/s

108. A ball falls from a height such that it strikes the floor of lift at 10 m/s. If lift is moving in the upward direction with a velocity 1 m/s, then velocity with which the ball rebounds after elastic collision will be

(a) 11 m/s (b) 12 m/s (c) 13 m/s (d) 9 m/s

109. A proton is moving with velocity u and collide with a nucleus of mass number A at rest then find fraction of retained K.E. if collision is elastic

110. A metal ball of mass 2 kg moving with speed of 36 km/h has a head on collision with a stationary ball of mass 3 kg. If after collision, both the balls move together, then the loss in K.E. due to collision is

(a) 100 J (b) 140 J (c) 40 J (d) 60 J

111. Two steel balls A and B of mass 10 kg and 10g rolls towards each other with 5 m/s and 1 m/s respectively on a smooth floor. After collision; with what speed B moves [perfectly elastic collision]?

(a) 8 m/s (b) 10 m/s (c) 11 m/s (d) Zero

112. Three objects A, B and C are kept in a straight line on a frictionless horizontal surface. The masses of A, B and C are m,m and 2m respectively. A moves towards B with speed of 9 m/s and makes collision in which coefficient of restitution is zero. Thereafter they makes a elastic collision with C. The final speed of C is $\frac{x}{2}$ m/s find x

113. A neutron makes a head on elastic collision with a stationary deuteron. The fractional energy loss of the neutron in the collision is :-

(a) $\frac{16}{82}$ (b) $\frac{8}{9}$ (c) $\frac{8}{27}$ (d) $\frac{2}{3}$

114. A particle of mass m_1 moves with velocity v_1 and collides with another particle at rest of equal mass. The velocity of the second particle after the elastic collision is

(a) $2v_1$ (b) v_1 (c) $-v_1$ (d) 0

115. Two identical balls A and B collide head-on elastically. If velocities of A and B, before the collision, are + 0.5 m/s and -0.3 m/s respectively then their velocities, after the collision, are respectively

(a) –0.5 m/s and +0.3 m/s (b) + 0.5 m/s and +0.3 m/s

(c) +0.3 m/s and –0.5 m/s (d) –0.3 m/s and + 0.5 m/s

116. Two ivory balls are placed together at rest. A third identical ball moving with velocity u in line of first two balls; as shown in figure collide head on elastically then



(a) Third ball comes to rest with second ball while first ball moves with speed u

(b) Third bail comes to rest and other two move together with speed $u/2$

(c) All three ball move together with speed $u/3$

(d) All three ball move in such a manner each makes angle 120° to each other

117. The coefficient of restitution e, for a perfectly elastic collision is

(a) 0 (b) -1 (c) 1 (d) ∞

118. A bullet weighing 10g and moving with a velocity 300 m/s strikes a 5 kg block of ice and drop dead. The ice block is kept on smooth surface. The speed of the block after the collision is

(a) 6 cm/s (b) 60 cm/s (c) 6 m/s (d) 0.6 cm/s

119. A ball moving with velocity 2 m/s collides head on with another stationary ball of double the mass. If the coefficient of restitution is 0.5 then their velocities (in m/s) after collision will be

(a) 0,2 (b) 0,1 (c) 1,1 (d) 1,0.5

120. A bullet of mass m moving with a velocity u strikes a block of mass M at rest and gets embedded in the block. The loss of kinetic energy in the impact is

(*a*) $\frac{1}{2}mMu^2$ (*b*) $\frac{1}{2}(m+M)u^2$

(*c*) $\frac{mMu^2}{2(m+M)}$ (*d*) $\left(\frac{m+M}{2mM}\right)u^2$

121. Two balls of equal mass undergo head on collision while each was moving with speed 6 m/s. If the coefficient of restitution is 1/3, the speed of each ball after impact will be

(*a*) 18 m/s (*b*) 2 m/s (*c*) 6 m/s (*d*) 4 m/s

122. An object of mass 80 kg moving with velocity 2 m/s^{-1} hit by collides with another object of mass 20 kg moving with velocity 4 m/s^{-1}. Find the loss of energy assuming a perfectly inelastic collision

(*a*) 12 J (*b*) 24 J (*c*) 30 J (*d*) 32 J

123. A body of mass 10 kg moving with speed of 3 m/s^{-1} collides with another stationary body of mass 5 kg. As a result, the two bodies stick together. The KE of composite mass will be

(*a*) 30 J (*b*) 60 J (*c*) 90 J (*d*) 120 J

124. Particle A makes a perfectly elastic collision with another particle B at rest. They fly apart in opposite direction with equal speeds. If their masses are m_A & m_B respectively, then

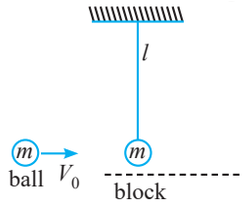
(*a*) $2\,m_A = m_B$ (*b*) $3\,m_A = m_B$

(*c*) $4\,m_A = m_B$ (*d*) $\sqrt{3}m_A = m_B$

125. A ball of mass M moving with speed v collides perfectly inelastically with another ball of mass m at rest. The magnitude of impulse imparted to the first ball is

(*a*) Mv (*b*) Mv^2 (*c*) $\frac{Mm}{M+m}v$ (*d*) $\frac{M^2}{M+m}v$

126. If collision is elastic then find maximum height attain by block just after collision from mean position??

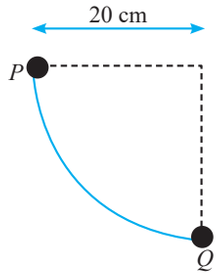


127. A bullet of mass m moving with velocity v strikes a suspended wooden block of mass M. If the block rises to height h, the initial velocity of the bullet will be

(*a*) $\sqrt{2gh}$ (*b*) $\frac{M+m}{m}\sqrt{2gh}$

(*c*) $\frac{m}{M+m}\sqrt{2gh}$ (*d*) $\frac{M+m}{M}\sqrt{2gh}$

128. As per the given figure, a small ball P slides down the quadrant of a circle and hits other ball Q of equal mass which is initially at rest. Neglecting the effect of friction and assume the collision to be elastic the velocity of ball Q after collision will be:



(*a*) 0.25 m/s (*b*) 4 m/s (*c*) 0 (*d*) 2 m/s

129. A ball is dropped from height h on the ground level. If the coefficient of restitution is e then the height upto which the ball will go after n^{th} jump will be:

(*a*) $\frac{h}{e^{2n}}$ (*b*) $\frac{e^{2n}}{h}$ (*c*) he^n (*d*) he^{2n}

130. A stone is dropped from a height h. It hits the ground with a certain momentum P. If the same stone is dropped from a height 100% more than the previous height, the momentum when it hits the ground will change by

(*a*) 68 % (*b*) 41% (*c*) 200% (*d*) 100%

131. A body of mass m falls from height h on ground. If e be the coefficient of restitution of collision between the body and ground, then the distance travelled by body before it comes to rest is

(*a*) $h\left\{\frac{1+e^2}{1-e^2}\right\}$ (*b*) $h\left\{\frac{1-e^2}{1+e^2}\right\}$

(*c*) $\frac{2eh}{1+e^2}$ (*d*) $\frac{2eh}{1-e^2}$

132. A body of mass m falls from height h on ground. If e be the coefficient of restitution of collision between the body and ground, then, total time taken by body before it comes to rest is

(*a*) $\sqrt{\frac{2H}{g}}\left\{\frac{1+e^2}{1-e^2}\right\}$ (*b*) $\sqrt{\frac{2H}{g}}\left\{\frac{1-e^2}{1+e^2}\right\}$

(*c*) $\sqrt{\frac{2H}{g}}\left\{\frac{1-e}{1+e}\right\}$ (*d*) $\sqrt{\frac{2H}{g}}\left\{\frac{1+e}{1-e}\right\}$

133. A rubber ball is dropped from a height of 5 m on a plane, where the acceleration due to gravity is not known. On bouncing, it rises to 1.8 m. The ball loses its velocity on bouncing by a factor of

(*a*) $\frac{3}{5}$ (*b*) $\frac{2}{5}$ (*c*) $\frac{16}{25}$ (*d*) $\frac{9}{25}$

134. A ball is allowed to fall from a height of 10 there is 40% loss of energy due to impact, after one impact ball will go up by

(*a*) 10 m (*b*) 8 m (*c*) 4 m (*d*) 6 m

135. A ball is dropped from a height of 20 m. If the ratio of velocity of separation to velocity of approach between ball and floor is 0.5. After hitting the floor, the ball rebounds to a height of____m.

136. A particle of mass m moving with a velocity $(3\hat{i}+2\hat{j})$ m/s collides with stationary body of mass M and finally moves with velocity $(-2\hat{i}+\hat{j})$ m/s then

(A) Impulse received by $m = m(5\hat{i}-5\hat{j})$

(B) Impulse received by $m = m(-5\hat{i}-\hat{j})$

(C) Impulse received by $M = m(5\hat{i}+\hat{j})$

(D) Impulse received by $M = m(5\hat{i}-\hat{j})$

then correct statements are

(*a*) B, C (*b*) A, C (*c*) B, D (*d*) A, D

137. A 1 kg ball falls from a height of 25 cm and rebounds upto a height of 9 *cm*. The coefficient of restitution is

(*a*) 0.6 (*b*) 0.32 (*c*) 0.40 (*d*) 0.56

138. A bullet of mass 10 g moving horizontally with a velocity of 400 m/s strikes a wooden block of mass 2 kg which is suspended by a light inextensible string of length 5 m. As a result, the centre of gravity the block is found to rise a vertical distance of 10 cm. The speed of the bullet after it emerges out horizontally from the block will be:

(*a*) 120 m/s (*b*) 160 m/s

(*c*) 100 m/s (*d*) 80 m/s

139. A moving block having mass *m*, collides with another stationary block having mass 4 m. The lighter block comes to rest after collision. When the initial velocity of the lighter block is *v*, then the value of coefficient of restitution (*e*) will be:

(*a*) 0.8 (*b*) 0.5

(*c*) 0.4 (*d*) 0.25

140. **Assertion (A):** Two particles moving in the same direction do not lose all their energy in a completely inelastic collision,

Reason (R): Principle of conservation of momentum holds true for all kinds of collisions.

(*a*) Both Assertion (A) and Reason (R) are correct and Reason (R) is correct explanation of Assertion (A)

(*b*) Both Assertion (A) and Reason (R) are correct and Reason (R) is not the correct explanation of Assertion (A)

(*c*) Assertion (A) is correct and Reason (R) is incorrect

(*d*) Reason (R) is correct and Assertion (A) is incorrect

141. Which of following statement is false

(*a*) KE in an elastic collision between two bodies remains same before and after collision

(*b*) There may or may not be physical contact between two bodies for collision to occur.

(*c*) KE of two object system remains conserved during collision in elastic collision

(*d*) None of the above

142. Particle of mass *m*, kinetic energy K and momentum P collides head on elastically with another particle of mass 2m at rest. After collision. Match column-I with column-II

Column-I		Column-II	
(A)	Momentum of 1st particle	P	$\frac{4}{3}P$
(B)	Momentum of 2nd particle	Q	$\frac{K}{9}$
(C)	KE of 1st particle	R	$\frac{8K}{9}$
(D)	KE of 2nd particle	S	$-\frac{P}{3}$

(*a*) A → S; B → P, C → Q, D → R

(*b*) A → P; B → S, C → Q, D → R

(*c*) A → P; B → S, C → R, D → Q

(*d*) A → S; B → P, C → R, D → Q

143. A particle of mass *m* with an initial velocity $u\hat{i}$ collides perfectly elastically with a mass 3 m at rest. It moves with a velocity $\hat{v}j$ after collision, then, *v* is given by

(*a*) $v=\frac{1}{\sqrt{6}}u$ (*b*) $v=\frac{u}{\sqrt{3}}$ (*c*) $v=\sqrt{\frac{2}{3}}u$ (*d*) $v=\frac{u}{\sqrt{2}}$

144. A ball of mass 200 g rests on a vertical post of height 20 m. A bullet of mass 10 g, travelling in horizontal direction, hits the centre of ball. After collision both travels independently. The ball hits the ground at distance 30 m and bullet at 120 m distance from the foot of post. The value of initial velocity of bullet will be:

(*a*) 120 m/s (*b*) 400 m/s

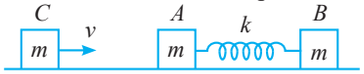
(*c*) 360 m/s (*d*) 60 m/s

145. What percentage of kinetic energy of a moving particle is transferred to a stationary particle when it strikes the stationary particle of 5 times its mass? (Assume the collision to be head-on elastic collision)

(*a*) 50% (*b*) 66.6%

(*c*) 55.6% (*d*) 33.3%

146. Two identical blocks *A* and *B* each of mass *m* resting on the smooth horizontal floor connected by a light spring of natural length *L* and spring constant *K*. *A* third block *C* of mass *m* moving with speed *v* along the line joining *A* and *B* slides with *A*. The maximum compression in spring is



(*a*) $\sqrt{\frac{mv}{k}}$ (*b*) $\sqrt{\frac{m}{2kv}}$

(*c*) $\sqrt{\frac{mv}{2k}}$ (*d*) $v\sqrt{\frac{m}{2k}}$

147. **Assertion (A):** Body '*P*' having mass *M* moving with speed '*u*' has head on collision elastically with Another body '*Q*' having mass '*m*' initially at rest. If $m << M$, body '*Q*' will have maximum speed equal to '2*u*' after collision.

Reason (R): During elastic collision, the momentum and kinetic energy are Both conserved

(*a*) Both Assertion (A) and Reason (R) are correct and Reason (R) is correct explanation of Assertion (A)

(*b*) Both Assertion (A) and Reason (R) are correct and Reason (R) is not the correct explanation of Assertion (A)

(*c*) Assertion (A) is correct and Reason (R) is incorrect

(*d*) Reason (R) is correct and Assertion (A) is incorrect

148. Body of mass 1 kg collides head on elastically with a stationary body of mass 3 kg After collision, the smaller body reverses its direction of motion and moves with speed of 2 m/s. The Initial speed of the smaller body before collision is _____ ms^1.

149. **Assertion (A):** Two particles moving in the same direction do not lose all their energy in a completely inelastic collision,

Reason (R): Principle of conservation of momentum holds true for all kinds of collisions.

(*a*) Both Assertion (A) and Reason (R) are correct and Reason (R) is correct explanation of Assertion (A)

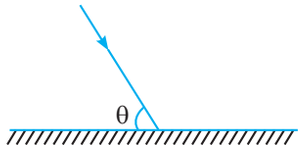
(*b*) Both Assertion (A) and Reason (R) are correct and Reason (R) is not the correct explanation of Assertion (A)

(*c*) Assertion (A) is correct and Reason (R) is incorrect

(*d*) Reason (R) is correct and Assertion (A) is incorrect

Oblique Collision

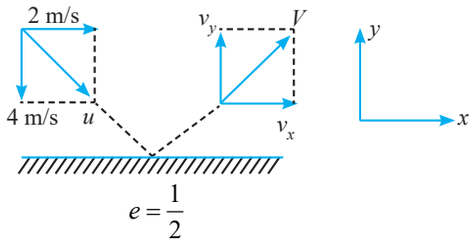
150. A ball of mass *m* moving with speed *u* collides with a smooth horizontal surface at angle θ with it as shown in figure. The magnitude of impulse imparted to surface by ball is [Coefficient of restitution of collision is *e*]



(*a*) $mu(1+e)\cos\theta$ (*b*) $mu(1-e)\sin\theta$

(*c*) $mu(1-e)\cos\theta$ (*d*) $mu(1+e)\sin\theta$

151. In the figure shown, a small ball hits obliquely a smooth and horizontal surface with speed *u* whose *x* and *y* components are indicated. If the coefficient of restitution is $\frac{1}{2}$, then its *x* and *y* components v_x and v_y just after 2 collision are respectively



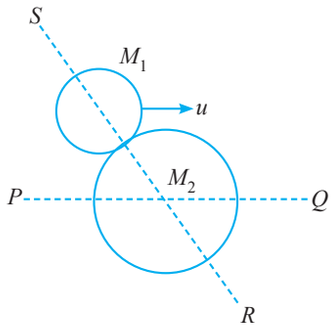
(*a*) 4 m/s, 1 m/s (*b*) 2 m/s, 1 m/s

(*c*) 2 m/s, 2 m/s (*d*) 4 m/s, 2 m/s

152. Two balls of masses *m* each are moving at right angle to each other with velocities 6 m/s and 8 m/s respectively. If collision between them is perfectly inelastic, the velocity of combined mass is

(*a*) 15 m/s (*b*) 10 m/s (*c*) 5 m/s (*d*) 2.5 m/s

153. An object of mass M_1 moving horizontally with speed *u* collides elastically with another object of mass M_2 at rest. Select correct statement.



(*a*) The momentum of system is conserved only in direction *PQ*

(*b*) Momentum of M_1 is conserved in direction perpendicular to SR

(*c*) Momentum of M_2 will change in direction normal to CR

(*d*) All of these

154. If a mass *m* collides elastically with another identical mass at rest in oblique manner, if direction of velocity of one ball after collision makes an angle 30° with horizontal direction, then angle made by another ball with horizontal direction will be

(*a*) 30° (*b*) 45°

(*c*) 60° (*d*) 90°

155. A particle of mass *m* moving eastward with a speed *v* collides with another particle of the same mass moving northwards with same speed *v*. The two particles coalesce on collision. The new particle of mass 2m will move with velocity

(*a*) $\frac{v}{2}$,North-East (*b*) $\frac{v}{\sqrt{2}}$, South-West

(*c*) $\frac{v}{2}$, North-West (*d*) $\frac{v}{\sqrt{2}}$, Noth-East

156. Two spheres A and B of masses m_1 and m_2 respectively collide. A is at rest initially and B is moving with velocity v along x-axis. After collision B has a velocity $\frac{v}{2}$ in a direction perpendicular to the original direction. The mass A moves after collision in the direction

(*a*) $\theta = \tan^{-1}\left(\frac{1}{2}\right)$ to the x-axis

(*b*) $\theta = \tan^{-1}\left(-\frac{1}{2}\right)$ to the x-axis

(*c*) Same as that of B

(*d*) Opposite to that of B

157. Two particles of equal mass m have respective initial velocities $\hat{i}$ and $u\left(\frac{\hat{i}+\hat{j}}{2}\right)$ They collide completely inelastically. The energy lost in process is:

(*a*) $\frac{1}{3}mu^2$

(*b*) $\frac{1}{8}mu^2$

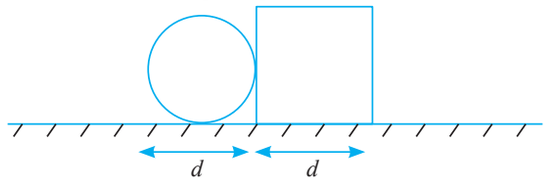
(*c*) $\frac{3}{4}mu^2$

(*d*) $\sqrt{\frac{2}{3}mu^2}$

MR* CORNER

1. **Statement I:** If All the particles of a Body are situated at a distance R from the origin. The distance of the Centre of mass of body from origin will be less than or equal to R.

   Statement II: A circular disc of diameter d is kept in contact with square plate of edge d as Shown. The density and Thickness being the same, The Centre of mass of system will be inside the square plate.



2. **Statement I:** Consider a system of Two identical particles. One of particles is at rest and other has an acceleration a The Centre of mass the accelerating particle. of mass will also have an acceleration a Because of accelerating particle.
   Statement II: Internal forces can change the kinetic energy but Not the linear Momentum.
3. **Statement I:** A uniform sphere is placed on smooth horizontal surface and horizontal force F is applied on it at a distance h above the surface. The acceleration of centre will be independent of h.
   Statement II: A Body falling vertically downwards under gravity breaks in two parts of unequal masses. The centre of mass of Two parts taken Together will shifts horizontally towards heavier piece.
4. **Statement I:** A Ball kept in a closed box moves in the box making collisions with walls. The Box is Kept on smooth surface. The Velocity of Centre of mass of box plus ball system will Remain constant.
   Statement II: A Body at Rest Breaks into Two pieces of different masses The parts will move in opposite direction with equal speeds.
5. **Statement I:** In an elastic collision the kinetic energy remains constant throughout.
   Statement II: If The centre of mass of system of particles is at origin, then The Total mass of particles to right of origin Must be same as Total mass to left of the origin.
6. **Statement I:** If the external forces acting on a system have zero Resultant, the Centre of mass must Not move.
   Statement II: If Nonzero external force acts on a system of particles. It is not impossible that velocity of c.o.m is zero But acceleration of c.o.m is non zero.
7. **Statement I:** If Two Balls are Thrown Simultaneously in air, Then acceleration of c.o.m of Two Balls while in air does depend on masses of Two Balls.
   Statement II: In an elastic collision of Two particles, the final linear momentum of each particle must be equal to their respective initial linear momentum.
8. **Statement I:** A Ball hits a floor and rebounds after an inelastic collision. In This Case The Total Momentum of Ball and earth is conserved.
   Statement II: A Body moving towards a finite body at Rest collides with it. It is possible that the stationary body remains stationary and the Moving body changes it's velocity.
9. **Statement I:** A Body Moving towards a finite body at Rest collides with it. It is possible that Both the Bodies come to Rest.
   Statement II: In a head-on elastic collision of Two bodies of equal masses, the fastest body slows dawn and the slower body speed ups.

Answer Key

1. $x_{cm_1} = \frac{m_2 r}{m_1 + m_2}, x_{cm_2} = \frac{m_1 r}{m_1 + m_2}$ **2.** (b) **3.** (a) **4.** (b) **5.** (c) **6.** (b) **7.** (c) **8.** (a)

9. (a) **10.** (a) **11.** $\frac{3\hat{i} + 10\hat{j}}{3}$ **12.** $\frac{-132}{15}$ **13.** $\frac{16\hat{i} + 20\hat{j}}{15}$ **14.** $\frac{6a\hat{i} + 8a\hat{j}}{11}$ **15.** $\frac{7b\hat{i} + 3\sqrt{3}b\hat{j}}{12}$

16. $m = 2$ **17.** $x = 2$ **18.** $2\hat{i} - 2\hat{j} + \frac{2}{3}\hat{k}$ **19.** (b) **20.** $\frac{18a}{15}$ **21.** (d) **22.** (b) **23.** (b)

24. $x = 3$cm **25.** (b) **26.** (b) **27.** (b) **28.** $\frac{L}{2\sqrt{2}}$ **29.** (b) **30.** (a) **31.** (a) **32.** (c) **33.** (a)

34. (c) **35.** (b) **36.** (a) **37.** $\left(\frac{7a}{6}, \frac{5a}{6}\right)$ **38.** 2R **39.** (b) **40.** 8 **41.** $\sqrt{3}R$ **42.** (b)

43. $x = \frac{R}{6}$ **44.** At Centre **45.** $\left(x = -\frac{R}{20}\right)$ **46.** $X_{com} = \frac{cb^2}{a^2 - b^2}$ **47.** $-\frac{3R}{56}\hat{i} - \frac{3R}{56}\hat{j}$ **48.** 2.25 m

49. $\frac{10\hat{i}}{3}$ m/s **50.** $\frac{14\hat{i} - 2\hat{j}}{3}$ **51.** (b) **52.** $\frac{5\sqrt{2}}{6}$ m/s^2 **53.** $\vec{a}_{cm} = \frac{-5\hat{i} + 2\hat{j} + 2\hat{k}}{4}$

54. $a = \left(\frac{m_2 - m_1}{m_2 + m_1}\right)^2 g$ **55.** (a) **56.** (b) **57.** (a) **58.** (b) **59.** Yes **60.** No **61.** Yes **62.** (b)

63. (a) **64.** (c) **65.** (a) **66.** (c) **67.** 13.3 cms **68.** (b) **69.** (d) **70.** (b)

71. 6 m/s **72.** (b) **73.** (d) **74.** (c) **75.** (b) **76.** (c) **77.** (b) **78.** (d) **79.** (d)

80. $V = \frac{5u}{3}$ **81.** (c) **82.** $\frac{K.E_1}{K.E_2} = \frac{m_2}{m_1}$ **83.** 2mv^2 **84.** $V_3 = -3\hat{i} - \hat{j}$ **85.** (c) **86.** (d) **87.** (a)

88. (b) **89.** (b) **90.** (c) **91.** $\frac{mL}{M + m}$ **92.** $\frac{1}{3}$m Towards right **93.** (b) **94.** (b) **95.** (b) **96.** (b)

97. (b) **98.** (a) **99.** (a) **100.** (a) **101.** (b) **102.** 12 **103.** (a) **104.** (d) **105.** (c)

106. 300 m/s **107.** (d) **108.** (b) **109.** $\left(\frac{1 - A}{1 + A}\right)^2$ **110.** (d) **111.** (c) **112.** $x = 9$ **113.** (b) **114.** (b)

115. (d) **116.** (a) **117.** (c) **118.** (b) **119.** (b) **120.** (c) **121.** (b) **122.** (d) **123.** (a) **124.** (b)

125. (c) **126.** $h = \frac{4m^2 V_0^2}{2g(M + m)^2}$ **127.** (b) **128.** (d) **129.** (d) **130.** (b) **131.** (a) **132.** (d) **133.** (b)

134. (d) **135.** 5 **136.** (a) **137.** (a) **138.** (a) **139.** (d) **140.** (a) **141.** (c) **142.** (a) **143.** (d)

144. (c) **145.** (c) **146.** (d) **147.** (a) **148.** 4 **149.** (a) **150.** (d) **151.** (c) **152.** (c) **153.** (b)

154. (c) **155.** (d) **156.** (b) **157.** (b)

MR* CORNER

1. Both are True **2.** I = False, II = True **3.** I = True, II = False **4.** I = True, II = False

5. Both are False **6.** I = False, II = True **7.** Both are False **8.** I = True, II = False

9. I = False, II = True